

OpenWebSearch.eu - Building an Open Index of the Web on HPC Infrastructure

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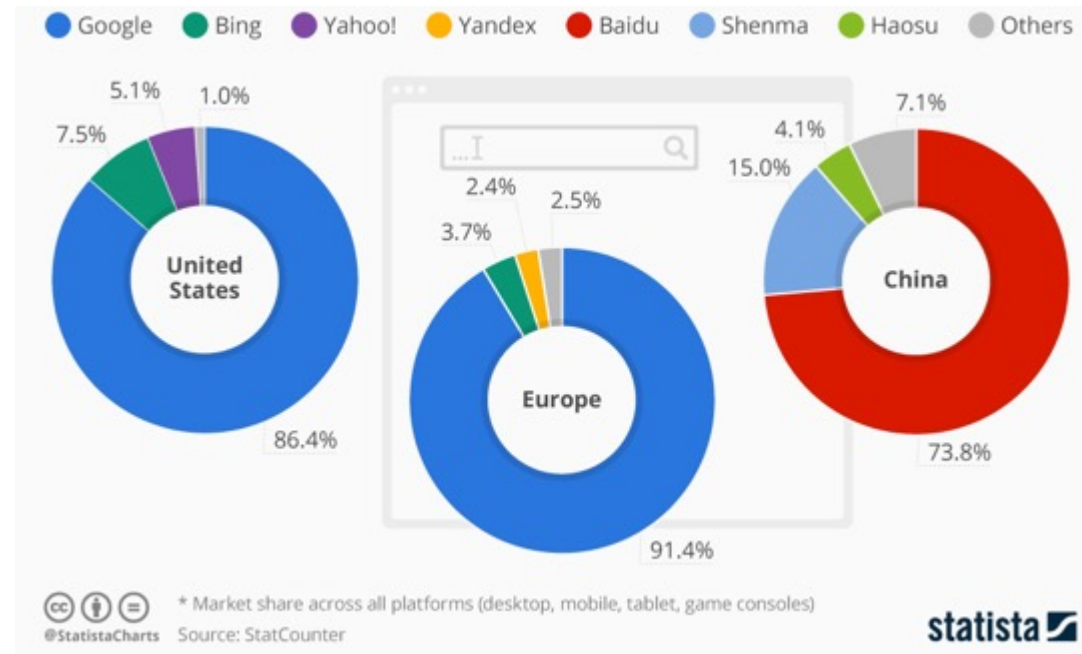
HPCSE 2026 · IT4Innovations · 2026-05-20

Web Search Today

A critical resource managed as an oligopoly

Two properties that don't fit

- A **critical infrastructure** for society, comparable to satellite navigation
- A **market oligopoly**: dominated by a small number of gatekeepers (Google, Microsoft, Baidu, ...)



But does it matter? Yes — on four dimensions

User Choice & Democracy

- Single gatekeeper for society's information
- Search rankings can shift voting preferences by **20–72%** (SEME)
- Monopolistic ranking = single point of failure for information quality

Untapped Richest Information Source

- 60 % of web pages carry structured data
- No open access to web-scale document collections
- Science, research and technology heavily depend on the Web as platform

Societal & Security Risks

- European OSINT depends on US-controlled infrastructure
- Filter bubbles, algorithmic bias
- Misinformation & Disinformation

AI & Innovation

- Web data is the fuel for LLMs, RAG, and agentic AI
- Closed indices block reproducible research & open innovation

Being able to navigate the Web is as fundamental as GPS or the power grid

Challenges

THE WEB: VOLUME, VELOCITY, VARIETY

Careful: domains, sites, pages, and indexed documents are different units

VOLUME

1.15 Billion

Netcraft site responses

272.6 M domains · Dec 2024 Netcraft survey

Top 0.1%–1% of sites

may account for roughly half of all pages
heavy-tail intuition from large web graph studies

~ 400 Billion

documents in Google's search index
2020 figure cited in US v. Google testimony

Sites ≠ domains ≠ pages; use exact counts cautiously

VELOCITY

+8.6 M / month

monthly net change in Netcraft site responses

Dec 2024 versus Nov 2024 Netcraft survey

~ 3.2 / second

same monthly net change, averaged across December

149 ZB (global)

all digital data created / copied worldwide in 2024
IDC / Statista global datasphere estimate

The web changes constantly; crawling is continuous, not one-off

VARIETY

150+ languages

observed across websites

W3Techs usage survey

249 CMS Platforms

CMS platforms identified in the HTTP Archive dataset
HTTP Archive Web Almanac 2024 (Wappalyzer-based)

HTML · PDF · JSON-LD · media · APIs

content types to parse and process

HTML · PDFs · JS apps · schemas · media · APIs

Sources: Netcraft Web Server Survey Dec 2024 · large web graph studies (heavy-tail intuition) ·
US v. Google testimony on ~400B documents (2020 figure) · HTTP Archive Web Almanac 2024 · W3Techs · IDC/Statista



- High infrastructure costs
- Broad range of technical skills required
 - Big Data Infrastructure
 - Natural Language Processing, Image Analysis
 - Web Technologies
- Growing legal uncertainties

Challenges

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- High infrastructure costs
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High upfront costs to tap the Web as resources — especially small innovators and researchers are left behind

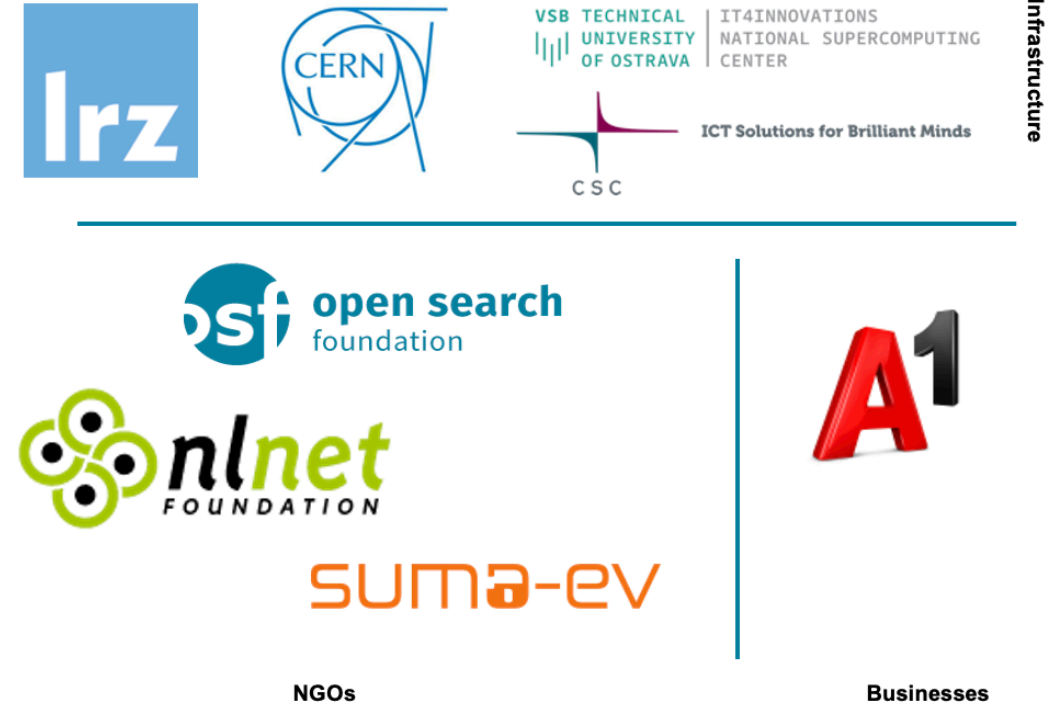
OpenWebSearch.eu — Our Mission

Building an Open Index of the Web and an open infrastructure to reduce upfront costs for researchers, innovators and companies in tapping the Web as a resource for search, web-analytics and AI.

Four Objectives

- 1. Open Technology Stack**
Open-source pipelines for crawling, preprocessing, indexing
- 2. Resource Provision** Building a network of infrastructure providers (EuroHPC / AI Factories)
- 3. Added Value Services**
Dashboard, tools, data access APIs
- 4. Bootstrapping the Ecosystem** Third-party calls, community building, applications and use-cases

14 Core Partners

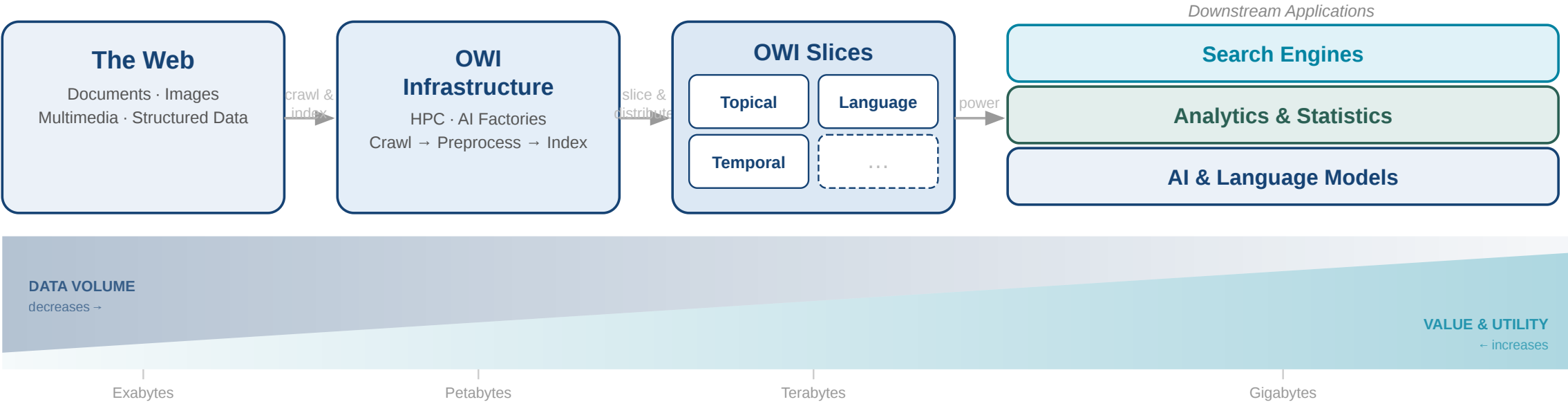


Plus 16 financially supported third-party projects (FSTPss)

The Open Web Index (OWI)

A **Web Index** = a data structure for fast query-based access and ranking of web documents — the core of every web search engine

Our Proposition: A collaboratively created, open and transparent Web Index running on existing HPC computing centres / AI Factories in Europe



Granitzer, Michael, et al. "Impact and development of an Open Web Index for open web search." *Journal of the Association for Information Science and Technology* (2023).

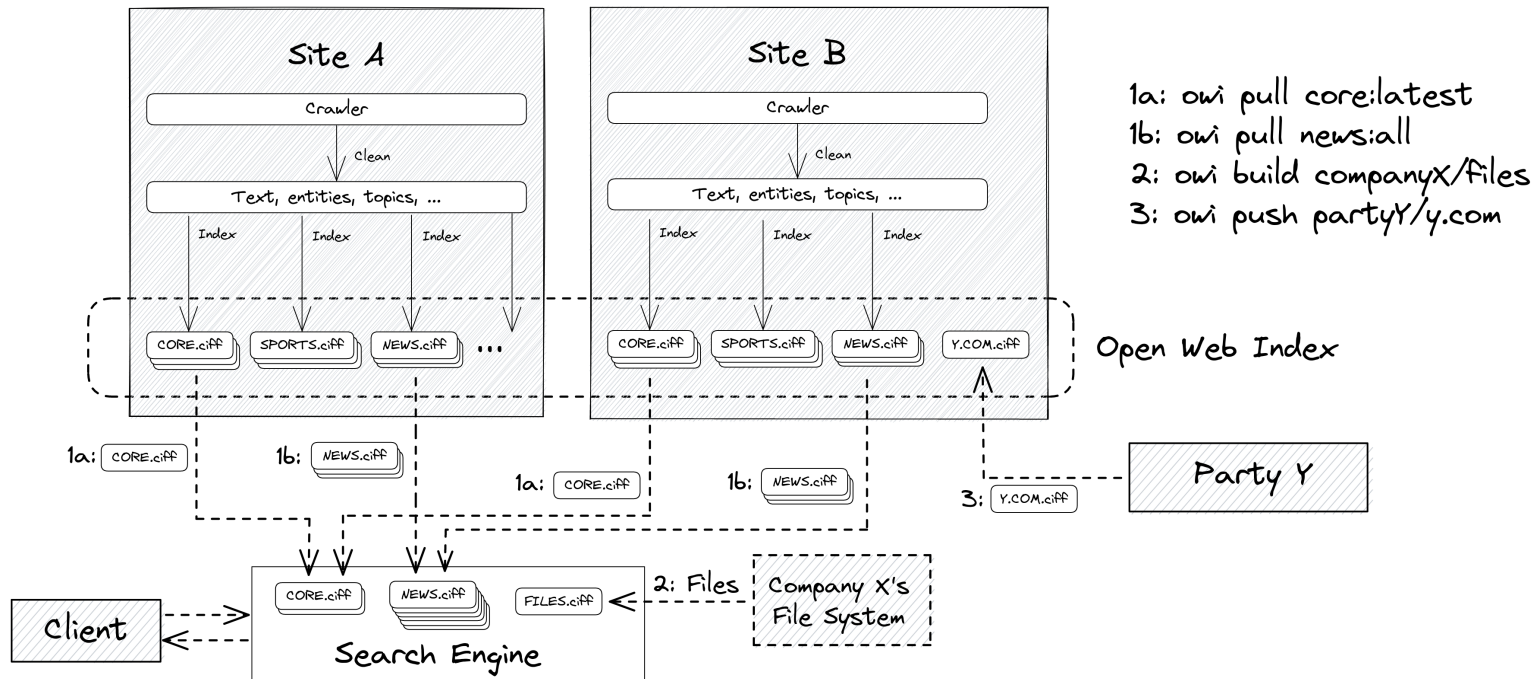
The Open Web Index

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The Open Web Index — a federated, daily-published web index built across European HPC sites and consumed as composable shards.

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Federated production

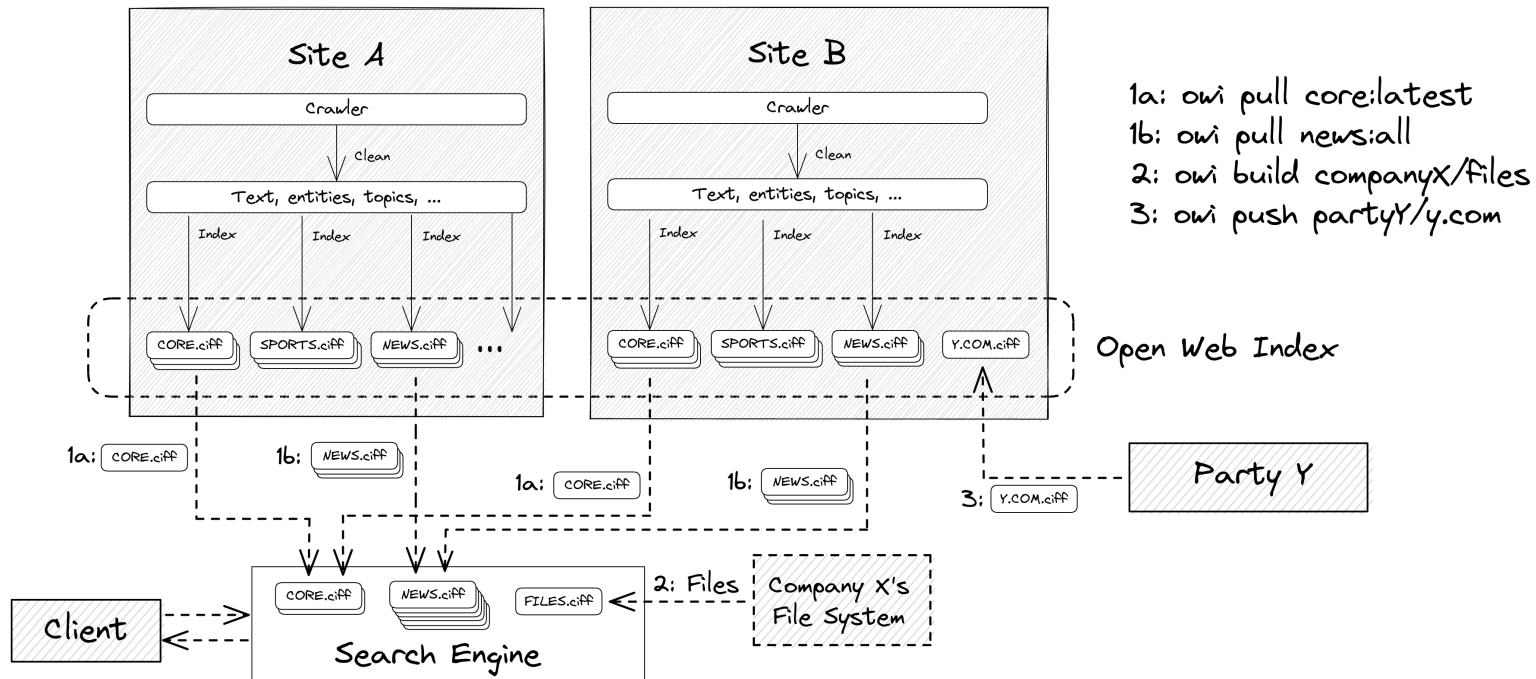
- each HPC site runs the same pipeline independently
- daily shards by date, language, topic

Three access verbs

- owi pull — download pre-built shards
- owi build — index your own data
- owi push — contribute shards back

The Open Web Index

The Open Web Index — a federated, daily-published web index built across European HPC sites and consumed as composable shards.



Federated production

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Three access verbs

- owi pull — download pre-built shards
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- owi push — contribute shards back

Three output types per daily run — the building blocks of a shard:

OWI at a Glance — Statistics (Feb 2026)

Crawling

Pages crawled	~39 B total, ~12 B unique
Unique hosts	~500 M (extrapolated)
Crawling effort	1,927 machine-days
Avg. throughput	~20 M URLs / day

Index

Documents indexed	~10.7 B
Unique domains	~75 M
Public datasets	511 main / 1,511 total
Index size	45 TB (main) / 50 TB total
Embeddings	119 M docs · 884 GB

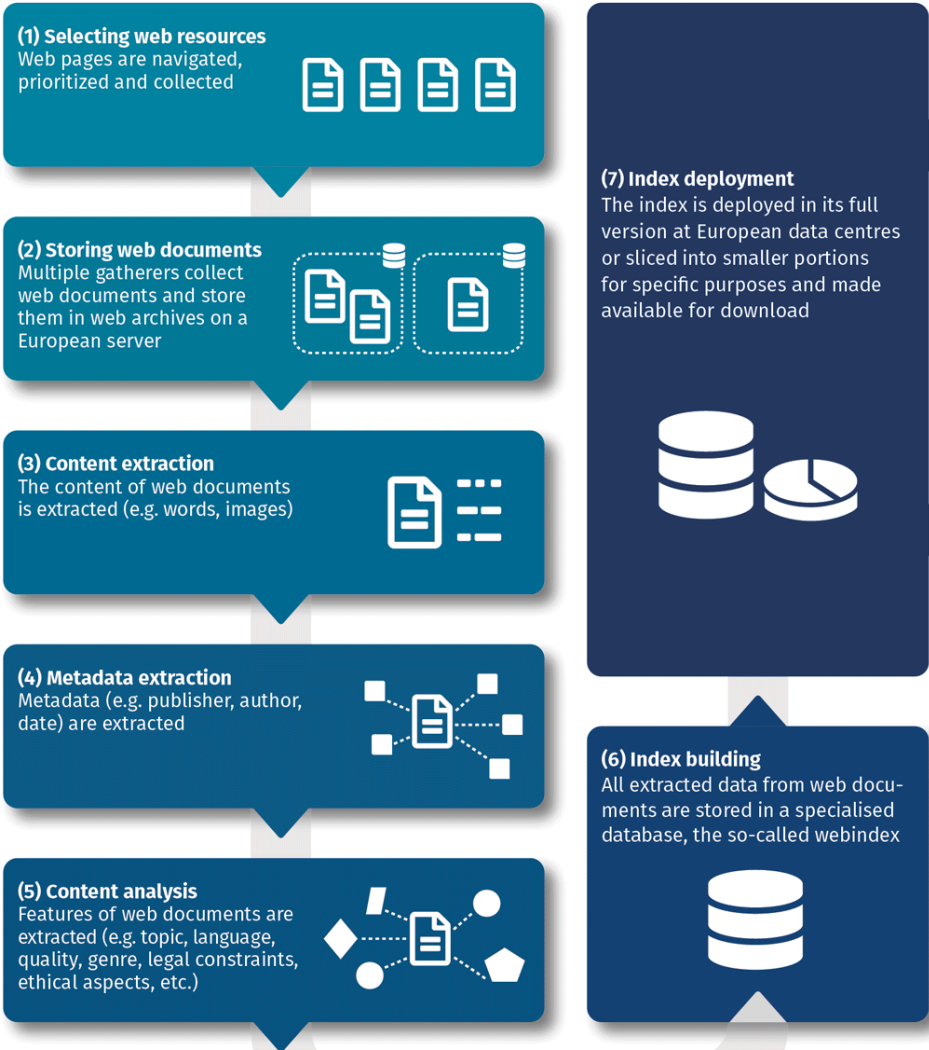
- Top languages:** English (38.9%), German (7.2%), Spanish (6.4%), French (5.7%), Russian (5.2%)
- Multimedia Objects:** Roughly 2-3 quality images and 0.5 Videos per Web-Document
- Comparison:** To meet commercial index size, OWI would need to scale by a factor of 20× — factor 10–50× for multimedia

Building the Open Web Index on HPC Infrastructure

Conceptual Workflow

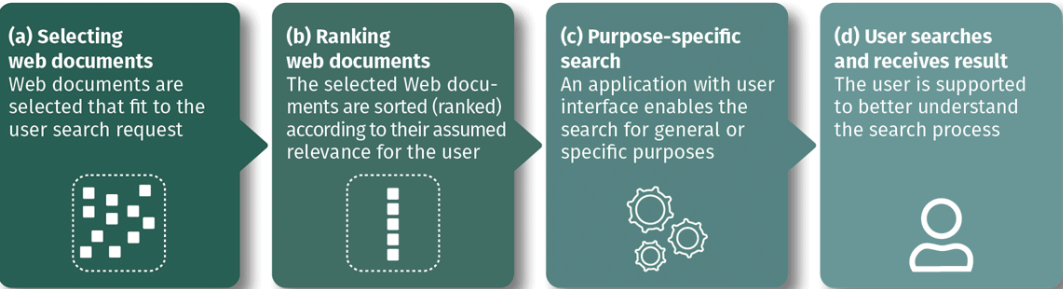
Index Generation

Web resources are selected and retrieved, their content and metadata are analysed, and all data stored in the index database.



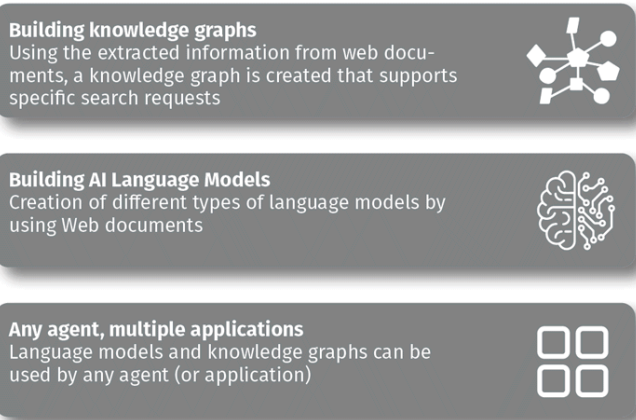
Search Applications

A user search request will be answered by a search application that makes use of the open web index.



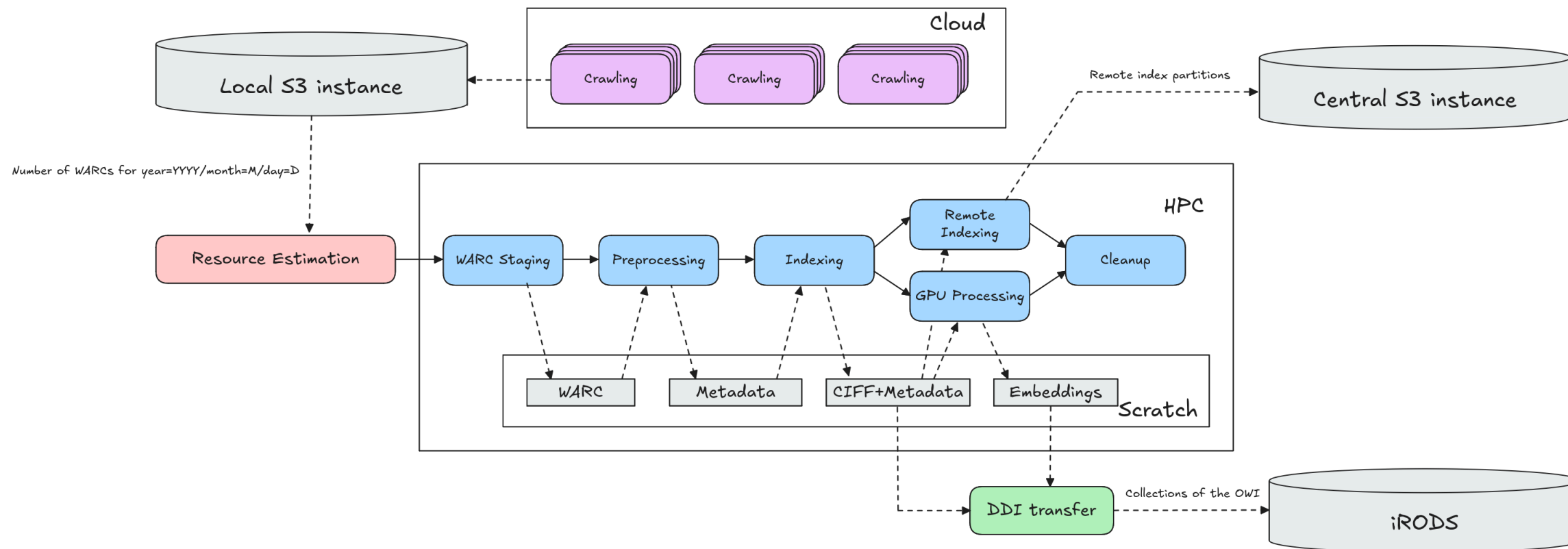
Data Products

Knowledge representation models will be created using the open web index, in order to be used by any agent and for many applications



...

Daily Workflow Pipeline



Crawlig Infrastructure (2026)

- **25 VMs** across IT4I, LRZ, CSC, CERN
- **100 M URLs / day** crawling capacity
- **~1.1 PB** raw data · **~50 TB** index · **810+ public datasets**
- Federated storage: iRODS + S3, orchestrated via LEXIS

Limitations as of 2026

- TLR 5/6: it is not a product (yet)
- No 24/7 Operations
- Limited funding - minimum engineering / improvement

The OWI as HPC Workload

A heterogeneous, recurring, web-scale workflow: each stage shifts the bottleneck, and the whole pipeline runs daily across federated sites. That combination only fits HPC.

Stage	Dominant pressure	Trend
Crawling	network, scheduling, politeness	stable
WARC processing	I/O, parsing, decompression	stable
Indexing	memory, sorting, CPU	stable
Embeddings	GPU throughput	↑ rising
Vector index	GPU memory, ANN compression	↑ rising

Why HPC, not commodity cloud

- network capacity for crawling at scale
- memory & CPU for daily index builds
- GPUs for semantic enrichment
- PB-class federated storage
- batch scheduling already in place at scientific centers

What is shifting

- GPU pressure rises as semantic enrichment becomes default
- vector indices must be scheduled next to classical indexing
- the HPC question is no longer only "can we index the web?" but "can we semantically enrich it often enough for AI search?"

No single commodity cluster covers all stages. HPC centers do — and turn the OWI into reusable public infrastructure rather than a private product.

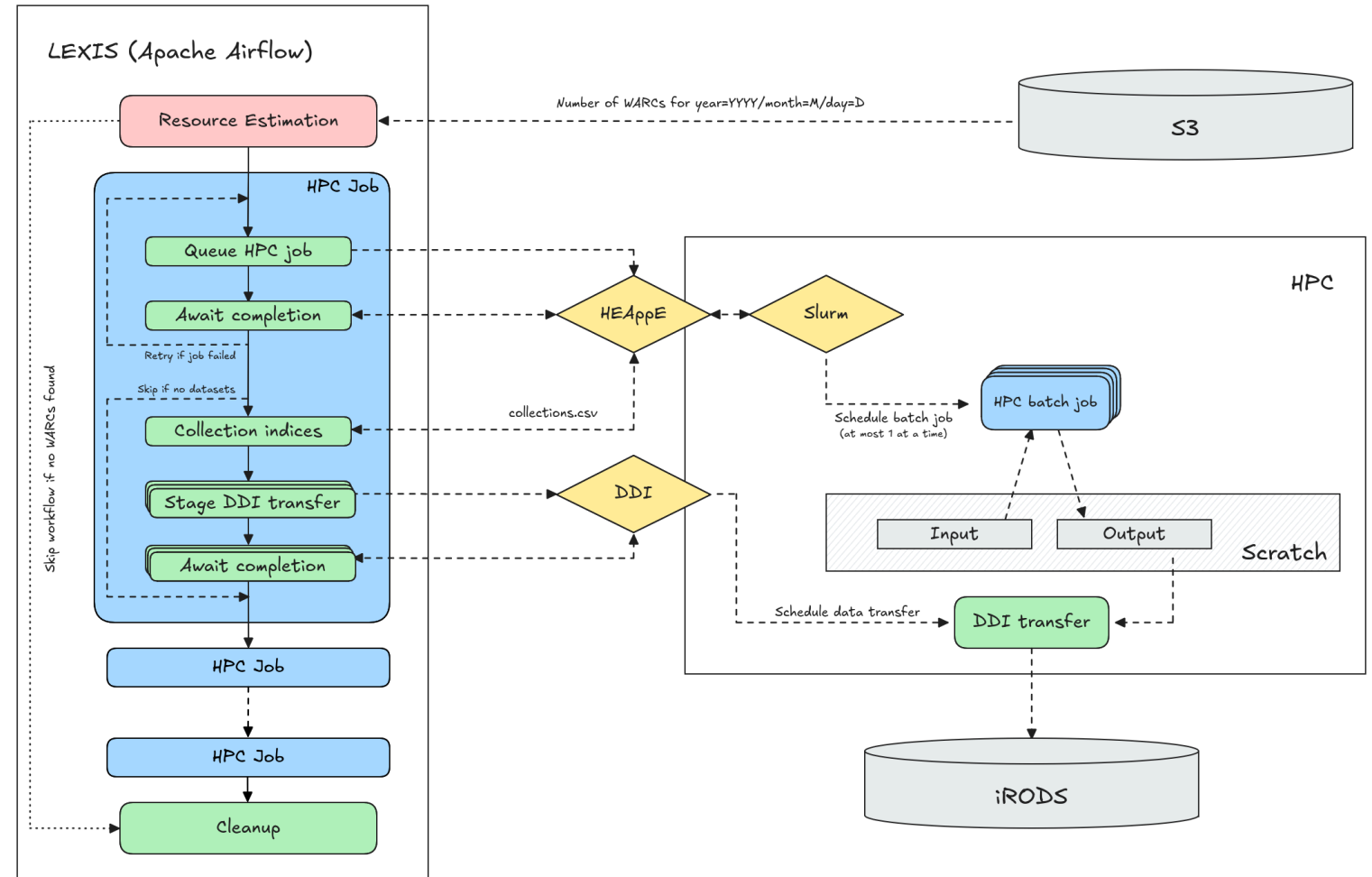
LEXIS: Orchestrating the Federation

The federation problem

- multiple HPC sites, each with its own scheduler, storage, identity
- the OWI must produce *aligned* daily output across all of them
- compute and data must meet — wherever the cycles happen to be

What LEXIS does

- coordinates OWI workflows across participating HPC / data centers
- stages input data and publishes daily OWI outputs
- bridges federated storage and compute jobs
- makes the build process repeatable



Workflow orchestration via LEXIS

LEXIS turns distributed HPC resources into a repeatable, federated Open Web Index build process — including data distribution across centers.

Accessing the Open Web Index

The Dashboard — openwebindex.eu

OWI

Dashboard

SERVICES

Websites

Search

SerCI Search EXTERNAL

Crawling

EXPERIMENTAL

Deep Research

UTILITIES

Parse Preview

Ethics Readiness Check

Impressum

Privacy Statement

Terms of Use

Open Web Search Book

Open WebSearch

Open Web Index

Overview OWI Statistics OWI Datasets OWI Corpora OWI Models

The Open Web Index

Your daily dose of web data!

Welcome to the Open Web Index (OWI)- your way to slice and dice the Petabytes in the Web to your needs. Our mission is to contribute to a fair, open, diverse and free Web by providing an open Index of the Web for you to use. This dashboard gives an overview over this [Open Web Index](#) and provides some prototype services on top of the OWI.



What can you find here?

Discover our comprehensive suite of tools and services designed to help you explore and analyze web data.



OWI Statistics
Get an overview of the data we offer



OWI Datasets
Browse through available datasets



OUR Search
Search through URLs and Titles



Documentation
Access tutorials and detailed documentation

Expect Bugs: While we try to provide as many valuable datasets and services as possible, please always remember that this is still a research project. So we cannot guarantee that everything works perfectly fine and we cannot take any responsibility if something is not working as intended. Bugs are to be expected, but if you encounter some, please get in touch with us so that we can improve.

Central entry point for the Open Web Index openwebindex.eu

- **OWI Statistics** — crawl volume, index size, language distribution, dataset timeline
- **OWI Datasets** — browse, filter and download index shards (by DC, date, format, language)
- **Search** — demo content search + SERCI integration

424 registered users · 17,000+ unique visitors · 70,000+ page views

Dashboard — Statistics & Datasets



- ### OWI Statistics (04/26)
- **1,389 TB** crawled · **281** WARC datasets · **1,696** public datasets · **35.09 TB** index
 - Language distribution chart (English 38%, German 8%, French 6%, ...)
 - Crawl volume over time per data center + dataset collection distribution by type
 - Datasets over time

Dashboard — Statistics & Datasets

Available Datasets

The data shows the datasets available for download (i.e. public owi datasets) or upon request (mostly project private warc datasets) by using our [LEXIS Platform](#) or our [OWI Command Line Tool](#). A dataset is a temporal slice, most often a single day, of crawled (warc), preprocessed and indexed data (owi).

All systems are online

All components are functioning properly. Data repositories and services are accessible.

Last Updated: 12/04/26, 15:36

SYSTEMS	DESCRIPTION	COMPONENTS			STATUS	
IT4I	IT4I	connected	Repository	Project	Public	ONLINE
LRZ	LRZ	connected	Repository	Project	Public	ONLINE

Filter datasets...

All Collections

All Datacenters

All Resources

Public

Sort by: Collection Date

OWI - Open Web Index

05/04/2026 → 05/04/2026

LEGAL

IT4I

PUBLIC

204.5 MB

253 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=fcb68e42-3191-11f1-8cbe-3e9ca2b0f43p

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

05/04/2026 → 05/04/2026

MAIN

IT4I

PUBLIC

24.3 GB

1,065 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=faba586e-3191-11f1-9f25-3e9ca2b0f43p

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

05/04/2026 → 05/04/2026

CURLIE_FULL

IT4I

PUBLIC

2.1 GB

365 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=fabf3050-3191-11f1-af7e-3e9ca2b0f43p

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

05/04/2026 → 05/04/2026

LICENSES

IT4I

PUBLIC

63.6 MB

163 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=fac99de2-3191-11f1-b11f-3e9ca2b0f43p

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

05/04/2026 → 05/04/2026

GPU

IT4I

PUBLIC

3.6 GB

374 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=faf9d060-3191-11f1-9f25-3e9ca2b0f43p

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

03/04/2026 → 03/04/2026

GPU

IT4I

PUBLIC

40.8 MB

6 files

All OWI data crawled between 2026-04-03 and 2026-04-03 at CSC2

owilix remote pull all/internalID=2f8f678c-2fd5-11f1-8ad8-3e9ca2b0f43p

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

03/04/2026 → 03/04/2026

LICENSES

IT4I

PUBLIC

275.8 KB

2 files

All OWI data crawled between 2026-04-03 and 2026-04-03 at CSC2

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

03/04/2026 → 03/04/2026

CURLIE_FULL

IT4I

PUBLIC

5.3 MB

8 files

All OWI data crawled between 2026-04-03 and 2026-04-03 at CSC2

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

03/04/2026 → 03/04/2026

MAIN

IT4I

PUBLIC

108.4 MB

73 files

All OWI data crawled between 2026-04-03 and 2026-04-03 at CSC2

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI Datasets Tab

- Status of repositories
- Filter by datacenter, format (WARC / Parquet / CIFF / Embeddings), language, date
- Dataset cards with download links — LEXIS Portal or owilix CLI
- Parquet file preview directly in the browser

20

Download Tool: owilix

owilix — The OWI Command-Line Tool

Git-like versioning for index datasets — pull, build, push:

```
owilix remote pull all:latest/collectionName=main
owilix local ls all:latest
```

Slice by: **domain**, **language**, **topic** (Curly), **sitelist**, **structured data**, **outlinks**
Authentication via B2ACCESS / Keycloak (LEXIS SSO)

Remote Index — Query first, then Download

- Parquet partitions directly queryable via **DuckDB** or **Apache Spark**
- Full-text search
- SQL access: filter by language, domain, date — no full-shard download needed
- Enables search + filtering + download (but still in alpha phase)

OWILIX - Open Web Index CLI

version 5.3.1

python 3.11+

license Apache 2.0

A command-line tool for managing [OpenWebIndex](#) datasets across local and remote repositories.

Quick Start

One-Line Install (Recommended)

```
curl -fsSL https://openwebindex.net/owilix/install.sh | sh
```

<https://opencode.it4i.eu/openwebsearcheu-public/owi-cli/>

```
➔ ~ owilix remote search "michael granitzer" --limit 10
Searching remote index: terms=['michael', 'granitzer'] language=eng representation=main_content mode=OR limit=10
Found 10 results in 113.4s
Search results (10 results in 113.4s)
```

score	date	url
3.088	2025-12-30	https://easychair.org/smart-program/ICADL2024/person16.html
3.010	2026-02-19	https://www.wolframalpha.com/input/?i=michael
3.010	2026-01-19	https://www.michaelmoennich.com/
3.010	2025-12-08	https://michaelkoenig.de/
3.010	2026-01-05	http://www.michaelmoerk.dk/
3.010	2026-01-22	https://michaelkoenig.de/
3.010	2026-02-18	https://www.sktthemes.org/forums/users/dedicationpt/replies/
3.010	2026-02-26	https://www.abeldiaz3.com/tag/michael/
3.010	2026-02-20	https://discourse.psychopy.org/u/michael
3.010	2026-02-27	https://thedigitalgobag.com/author/michael/

The OpenWebSearch Book

Comprehensive Documentation at openwebindex.eu/book

- Covers the full OWI ecosystem:
- Getting started with owilix
 - Building vertical search engines with MOSAIC
 - Data formats: CIFF, Parquet, embeddings
 - Pipeline architecture and HPC deployment
 - Legal & ethical guidelines for index users
- Published as an open, continuously updated online resource — aligned with deliverables and community feedback.

Access Modes at a Glance

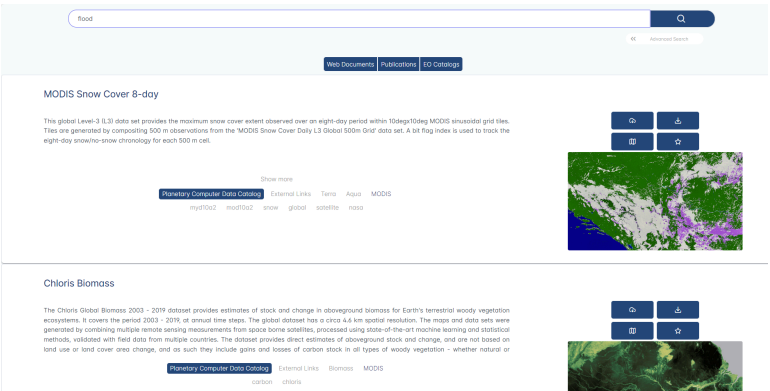
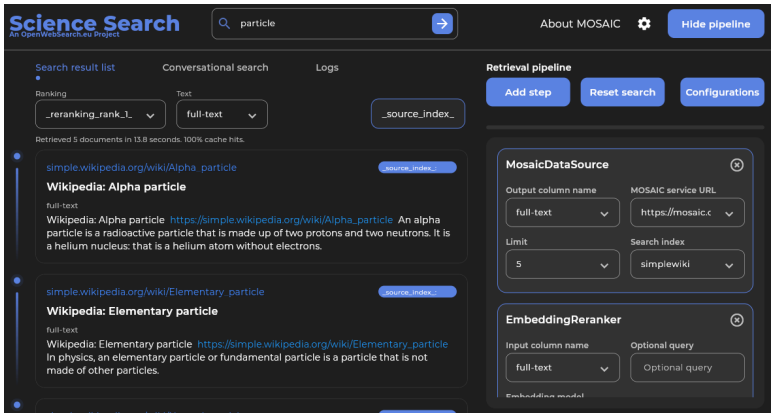
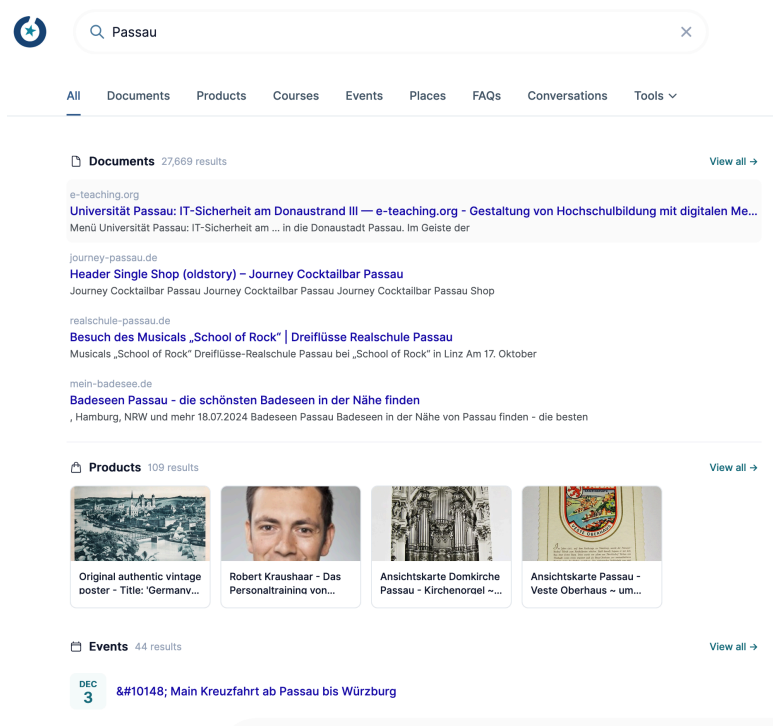
Tool	Mode	Best for
Dashboard	Browser	Explore, monitor
owilix	CLI	Download, slice, build
Remote index	DuckDB/SQL	Query + download
MosaicRAG	Conversational	RAG & chat over OWI

All software, data formats and access tools are **open source**



Using the Open Web Index

Using the Open Web Index — Search and RAG



OUR²S
Hosted search UI and REST API for
online retrieval directly over OWI
content.

Mosaic vertical search
Domain-specific search engines for
science, health, arts, and other focused
collections.

Earth observation search
RAG-based access to scientific papers,
datasets, and geo-contextualized web
content.

The OWI is useful when the application needs inspectable web-scale source material, not only a black-box answer.

Using the Open Web Index — Domain Applications

Energy-centric optimization of large-scale additive manufacturing deployment for sustainability through electrification

Rahoituksen hankkeen kuvaus

The employment of additive manufacturing (AM), commonly called 3D printing, in the industrial sector has been growing for many years. As the technology matures, it is expected that its use will become widespread and substitute other established forms of production. This project is...

Näytä enemmän

Aloitusvuosi

2026

Päättymisvuosi

2029

Myönnetty rahoitus

Humberto Alameda Jantar

Lappeenranta-Lahden teknillinen yliopisto LUT

305 350 €

Ruohi Suomen Akatemian konsertissa

Partneri

Muut osapuolet

Partneri

Aalto-yliopisto (372725)

296 731 €

Johdaja

Lappeenranta-Lahden teknillinen yliopisto LUT (372723)

395 919 €

Rahoittaja

Suomen Akatemia

Rahoitusmuoto

Suunnattu akatemiahanke

Haku

2025 Tulevaisuuden kestävien energiatekniikoiden tutkimuksen haku 2025

Muut tiedot

Rahoitusvälikkeen numero

372724

Tutkimusalat

Kone- ja valmistustekniikka

Tutkimusalat

Kone- ja valmistustekniikka

Myöntöön liittynyt verkko

Rahoitusmyöntöön ei ole tarjolla linkkejä tässä portaalissa

Vittaukset muissa palveluissa

Vittauksia yhteensä: 20

40%

(8)

30%

(6)

20%

(4)

10%

(2)

0%

(0)

Hakutulokset 1 - 5

https://www.industryweek.com/...

3 Ways Additive is Shaping the Future of the Industry

https://www.industryweek.com/...

These industries are the Future of Additive Manufacturing | IndustryWeek

https://global-ai.com/...

Digital Science Thesis: A Review of Process Optimization for Additive Manufacturing Based on Machine Learning

https://www.engineering.com/...

Customized Production 100 Rooms Studable with 3D Printing - Engineering.com

https://www.asea.org/...

Reprints for the Next Available Editors of ACS Publications Journals: Q3 2018 | ACS Publications Chemistry Blog

args

We should abolish the right to keep and bear arms

violence

520 results · 46 ms

All

Search

Pro-Arguments

Argumentative Segment

Since the start of the year, there have been 80 mass shootings and more than 3,300 deaths due to gun **violence** nationwide, according to the Gun **Violence** Archive. We don't need any more thoughts and prayers from elected leaders. We shouldn't have any more moments of **silence** or vigils. We need action.

Key Point · Quality Score: 0.880

Gun ownership allows for mass-shootings/general gun violence

Con-Arguments

Argumentative Segment

Councilman Johnson, repeat after me: **Violence is Violence**. The number one cause of **violence** is **violent** people. While semantically there is an argument that some people aren't **violent** until provoked, drunk and/or stoned, exposed to constant **violence** in entertainment media, video games etc., the fact remains that **violence** always, 100% of the time, involves at least one **violent** person. Blaming guns for **violence** is like blaming hammers for shoddy home construction.

Key Point · Quality Score: 0.660

Argumentative Segment

"Any successful strategy to reduce gun **violence** requires preventing the diversion of lawful firearms into unlawful commerce." Steven Dettelbach said at the time. "Once these firearms end up in the hands of people who are sometimes **violent** criminals and intend to do harm to the people with whom we live, the innocent people who are victims and survivors of gun **violence**."

Key Point · Quality Score: 0.690

Guns can fall into the wrong hands

Argumentative Segment

No matter how many times actual, real-life case studies are laid out for Cleveland's elected officials, they refuse to recognize that the only PROVEN method of reducing **violence** is removing **violent** persons from circulation, either through imprisonment or by vibrantly empowering the potential victims to remove the **violent** person when confronted by one.

Key Point · Quality Score: 0.870

Gun ownership promotes self protection

MOSAIC

ows-lb-2.cern.ch/Ui/

80%

MOSAIC Search

Search term: disability

Search

Index info

Geo Filter: West: East: North: South:

Index: default / all disability-index

Language: default / all English German

Limit: 20 10 Items 50 Items 1,000,000

Keyword:

Search URL: https://ows-lb-2.cern.ch/mosaic/search?q=disability&index=disability-index

Search result for term: "disability"

Index: disability-index

Number of items: 20

Cockrell Hill Social Security Disability Attorneys | OneLife Texas Attorney Directory

Kraft Dallas, TX Social Security Disability Lawyer with 54 years of experience (214) 999-9999 2777 N Stemmons Fwy Suite 1300 Dallas, TX 75207 Free Consultation Social Security Disability and Personal Injury Baylor Law School Show Preview View Website

Metadata: language:eng, word count:4273, index date:2025-08-10 00:13

Locations:

Keywords: Attorney · Attorneys · Social Security Disability · Cockrell Hill · Texas · https://lawyers.onecle.com/lawyers/social-security-disability/texas/cockrell-hill Insurance Coordinator Manual

Partial disability must result from the same condition as the total disability. Proof of partial disability must be submitted within 31 days of the date the employee's total disability period ends. Pa

Metadata: language:eng, word count:20858, index date:2025-08-10 00:51

Locations:

Keywords:

https://oklahoma.gov/legid/insurance-coordinators/manual.html

House of Lords/Commons - Joint Committee on the Draft Disability Discrimination Bill - First Report

Disability Rights Commission, DDB 1, para 10.6.1 British Council of Disabled People, DDB 6, para 9. Law Society, DDB 15, p.7. 15(a): The Government agrees with the recommendations of the DRTF that MPs

Metadata: language:eng, word count:12308, index date:2025-08-10 00:25

Locations:

Keywords:

https://publications.parliament.uk/pa/r200304/rlselect/jdisab/R2/R222.htm

UKRI diversity data for funding applicants and awardees 2020 to 21 update – UKRI

After removing this opportunity, the award rate for fellows reporting a disability was 22% compared with

Research.fi

National research discovery enriched with OWI-derived scientific web content.

Restaurant recommender

Privacy-preserving local recommendation from extracted restaurant features.

Argumentation search

Pro/con argument retrieval over curated OWI slices.

Institutional search

CERN institutional search and disability knowledge search.

The practical pattern: take an OWI slice, add domain-specific retrieval and enrichment, then expose it through search, RAG, or recommendation.

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Using the Open Web Index - Data for AI and LLMs

Using the Open Web Index — Data & Analytics

Published Datasets

Dataset	Scale	Use case
WebFAQ	96M Q&A, 75 langs	QA training, cross-lingual IR
WebFAQ 2.0	198M pairs, 108 langs	Dense retrieval, hard negatives
Imprints	5.25M hosts, geocoded	Enterprise analytics, Eurostat
German Commons	154B tokens	German LLM pre-training
OWS-Curlie-2025	20.7M docs + qrels	IR evaluation benchmark

All datasets available at openwebindex.eu/corpora

Analytics Potential

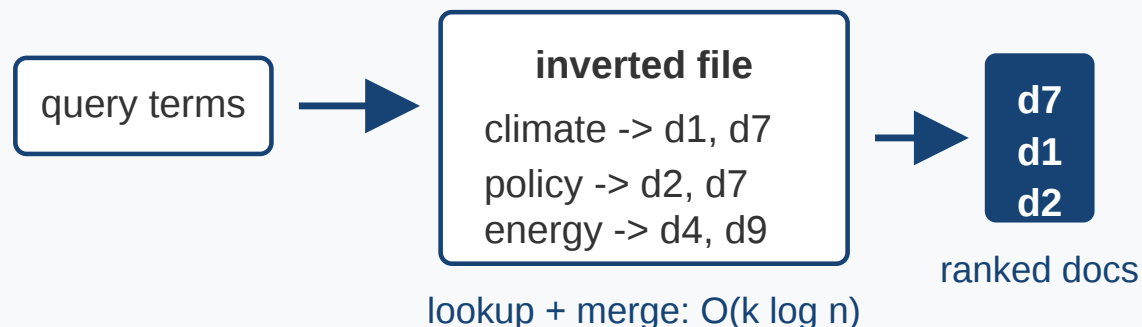
- **Official Statistics:** enterprise data extraction from legal imprints (Eurostat, Destatis)
- **Media Monitoring:** supply chain analysis, news monitoring (JRC)
- **AI Training Data:** domain-specific corpora for LLM fine-tuning
- **Evaluation Benchmarks:** web-scale IR test collections with relevance assessments
- **Trend Analysis:** temporal web monitoring at Petabyte scale

The OWI is not just for search — it is a data infrastructure for the web-scale analytics ecosystem

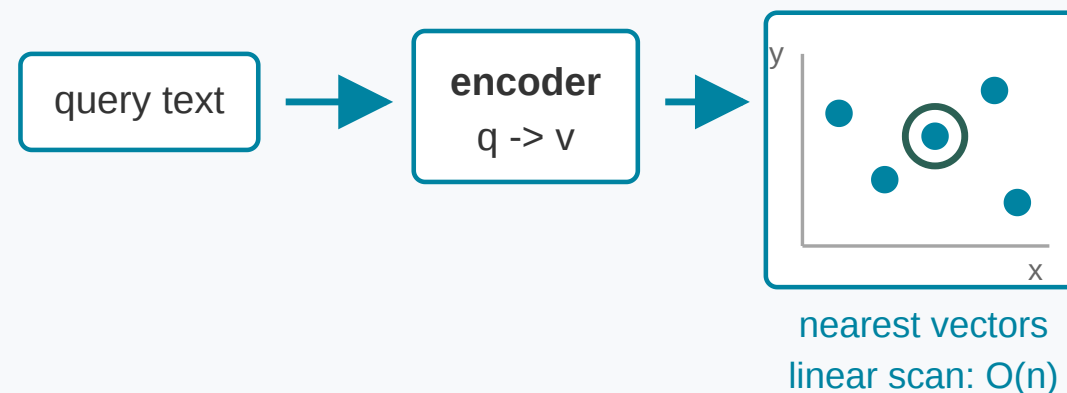
From Search to AI-Based Search

Retrieval Modes on the Open Web Index

Sparse index



Dense index



Sparse search

- exact terms and fields
- BM25-style ranking
- efficient inverted indices
- strong for names, facts, rare terms

Dense search

- semantic similarity in vector space
- robust to paraphrases
- useful for RAG and question answering
- needs vector storage and fast ANN

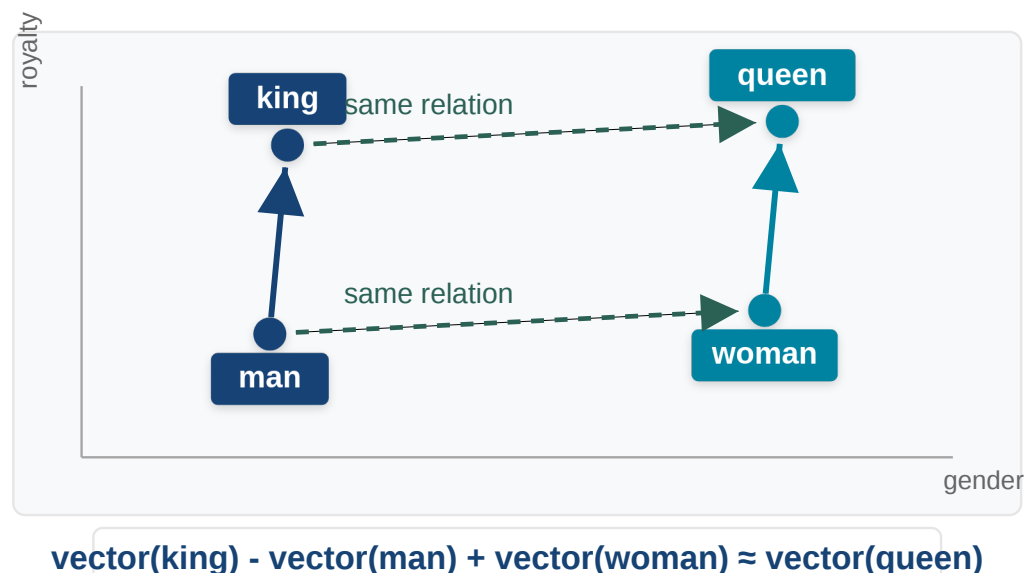
Hybrid search

- combine lexical and semantic evidence
- use metadata filters
- rerank with embeddings or LLMs
- balance precision and recall

≡ The IR progression: lexical matching gives efficient candidates; embeddings add semantic matching; hybrid search

combines both and raises the serving question.

Embeddings Encode Relations



Analogy example after Mikolov et al., *Efficient Estimation of Word Representations in Vector Space* (Word2Vec), [arXiv:1301.3781](https://arxiv.org/abs/1301.3781).

The classical word embedding example illustrates the intuition:

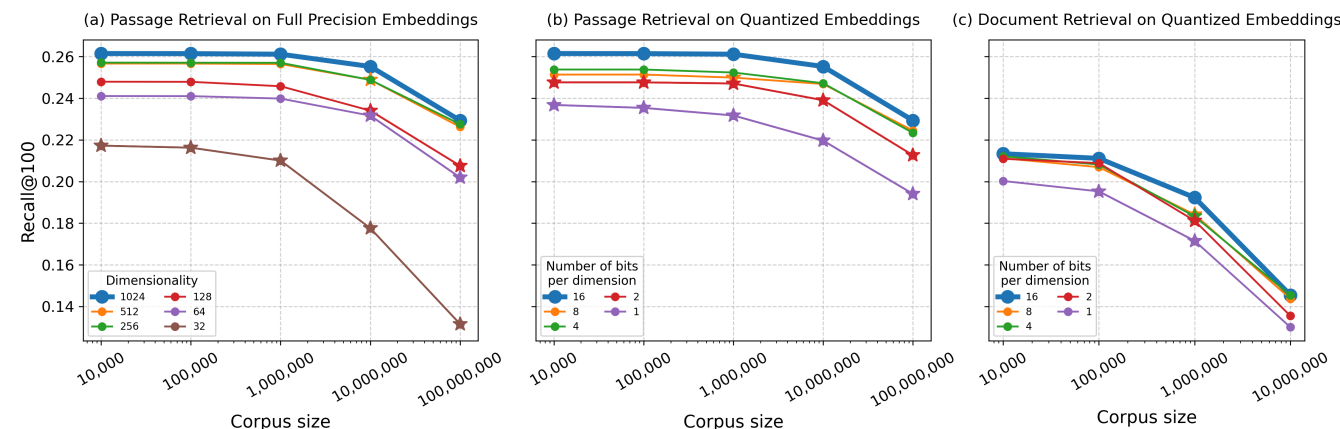
$$\text{king} - \text{man} + \text{woman} \approx \text{queen}$$

Embeddings place objects in a vector space where some semantic relations become approximately geometric operations.

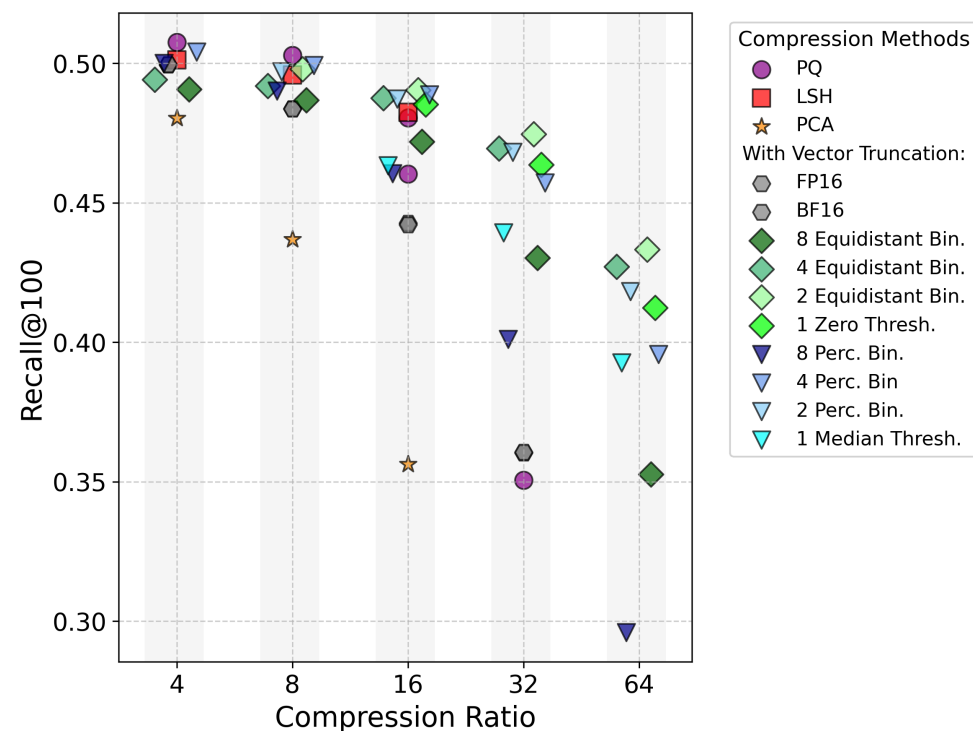
For search this creates a systems question: low-latency vector retrieval needs large vector indexes, often RAM-resident or close to RAM, plus substantial persistent storage.

Order of magnitude: at 1024-dim float32, 1 M vectors $\approx$ **4 GB** and 100 M vectors $\approx$ **400 GB**. The OWI currently ships **119 M dense vectors at ~884 GB**.

CoRECT: Evaluating Embedding Compression at Scale



Jina v3, CoRECT line chart, NDCG@10. Source: padas-lab-de.github.io/CoRECT



Jina v3, CoRECT Pareto plot, NDCG@10. Source: padas-lab-de.github.io/CoRECT

CoRECT

- Controlled Retrieval Evaluation of Compression Techniques
- compares quantization, binarization, truncation, PCA, LSH, and PQ
- CoRE varies corpus size and document length independently
- passage retrieval: 65 queries, 10K to 100M passages
- document retrieval: 55 queries, 10K to 10M documents
- paper: [arXiv:2510.19340](https://arxiv.org/abs/2510.19340)

CoRECT: Compression Trade-Offs

Results

- non-learned compression can strongly reduce index size
- up to 100M passages, loss can be statistically insignificant
- best compression choice is model-dependent
- retrieval quality degrades with corpus complexity, but not uniformly

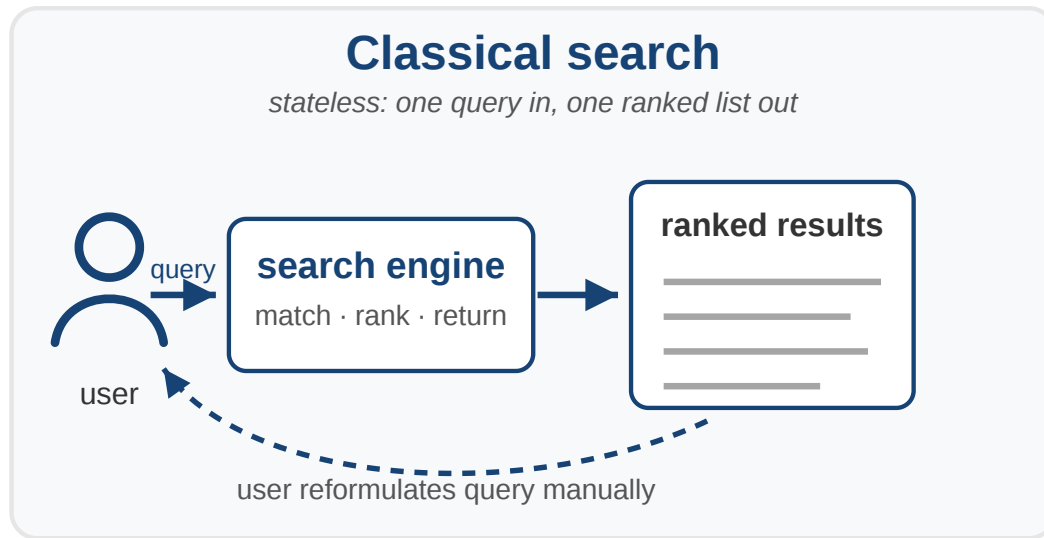
Serving conclusion

- uncompressed dense indexes are expensive to keep fast
- FP casting, scalar quantization, truncation, binarization, LSH, PCA, and PQ occupy different cost-quality regions
- the best point is not universal; it depends on model, metric, and corpus complexity
- open theoretical question: which ranking families can binary vectors express under Hamming distance or binary dot product?

Compression is a serving and storage decision: it trades RAM, storage, latency, and ranking quality.

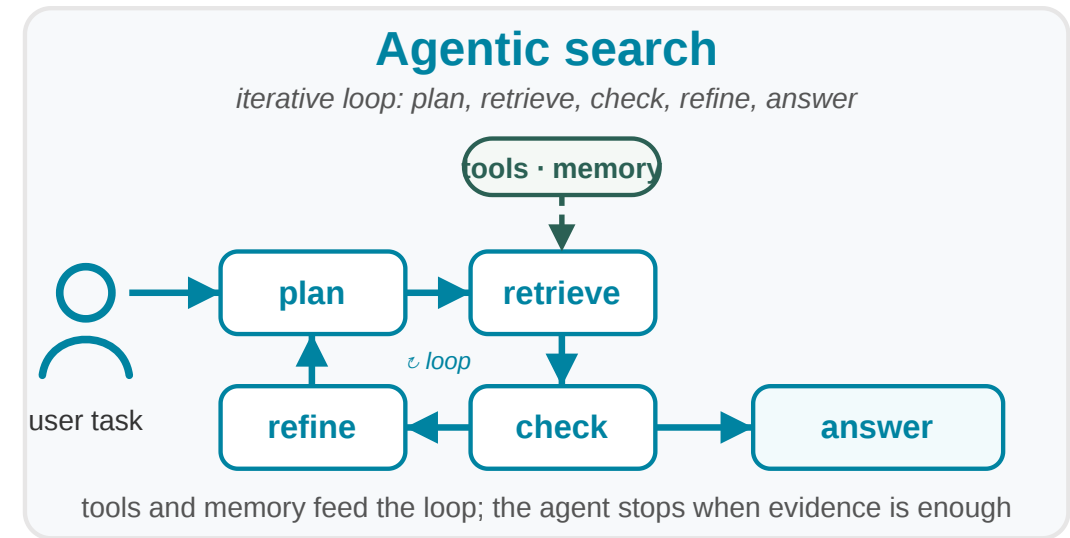
Agentic Search

From Search Engine to Search Agent



Classical search

- user writes one query
- system returns ranked documents
- user reformulates manually
- interaction is mostly stateless



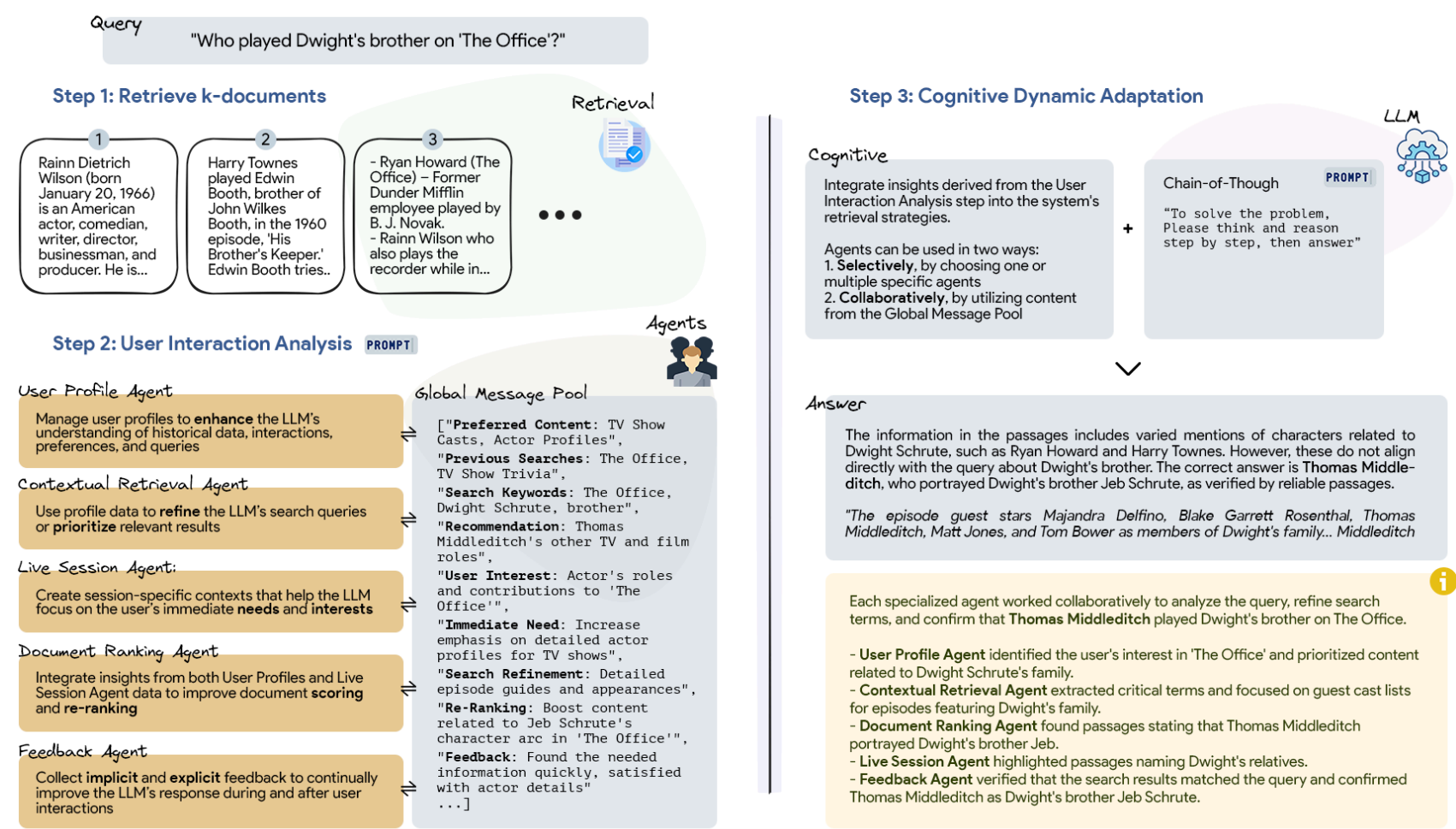
Agentic search

- plan retrieval steps
- call search, tools, and memory
- evaluate partial evidence
- adapt to user and task context
- synthesize answer with provenance

Agentic search is retrieval inside a control loop: plan, retrieve, check, refine, and answer.

PersonaRAG: Retrieval with User-Centric Agents

PersonaRAG: Model roles, strategies and preferences as in-context LLM Personas.



Design tension

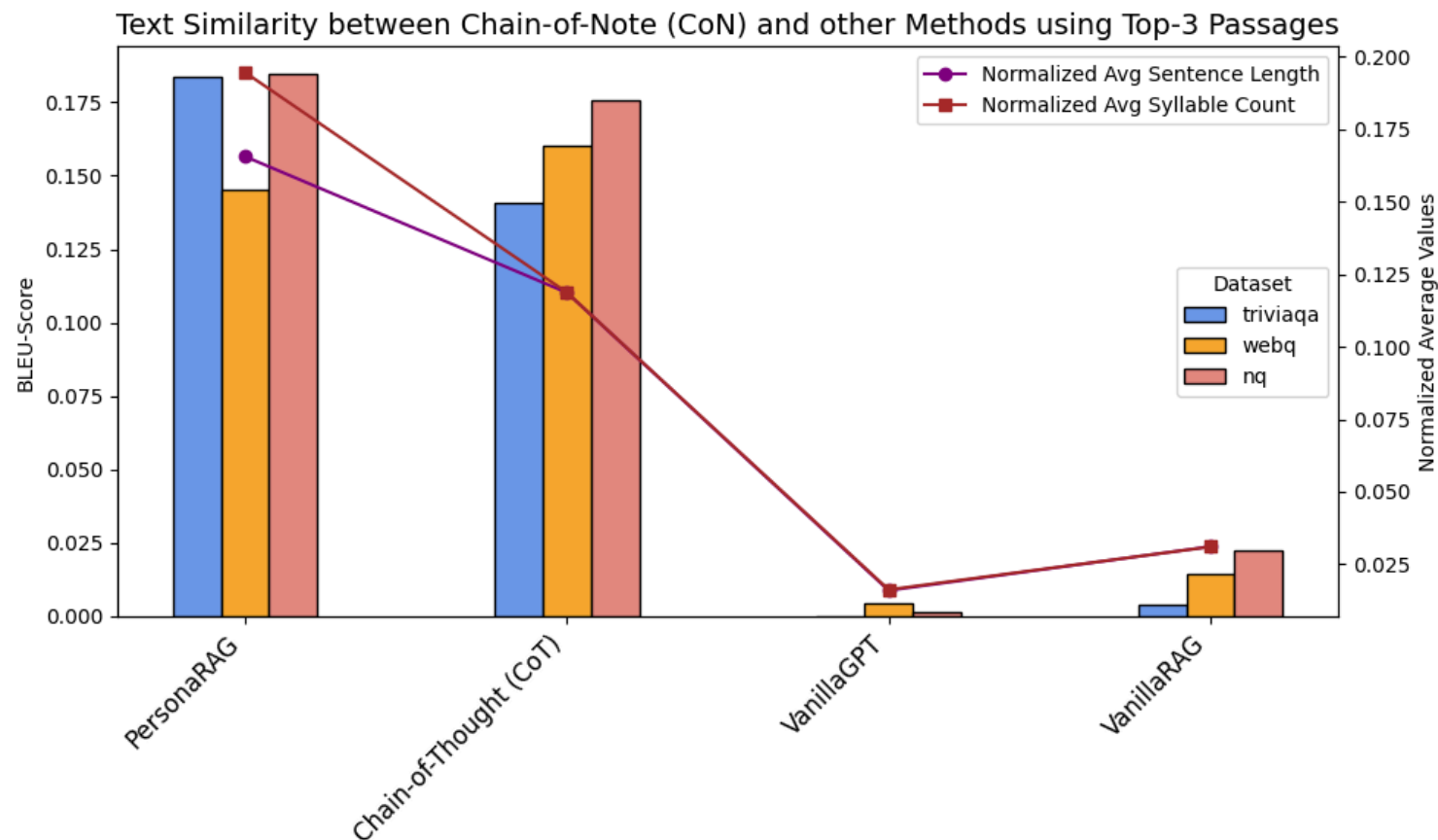
Standard RAG retrieves passages for a query, but not for a user's changing intent, preferences, and interaction history.

Mechanism

- user profile agent
- contextual retrieval agent
- live session agent
- document ranking agent
- feedback agent

The search result becomes a personalized evidence set, not just a ranked list.

PersonaRAG: Evaluation and Findings



Text similarity experiment from PersonaRAG, [arXiv:2407.09394](https://arxiv.org/abs/2407.09394)

Conclusion for agentic search: personalization improves answers, but it also makes context management a first-class retrieval problem — what to keep, update, forget, explain, and expose to the agent.

How it was tested

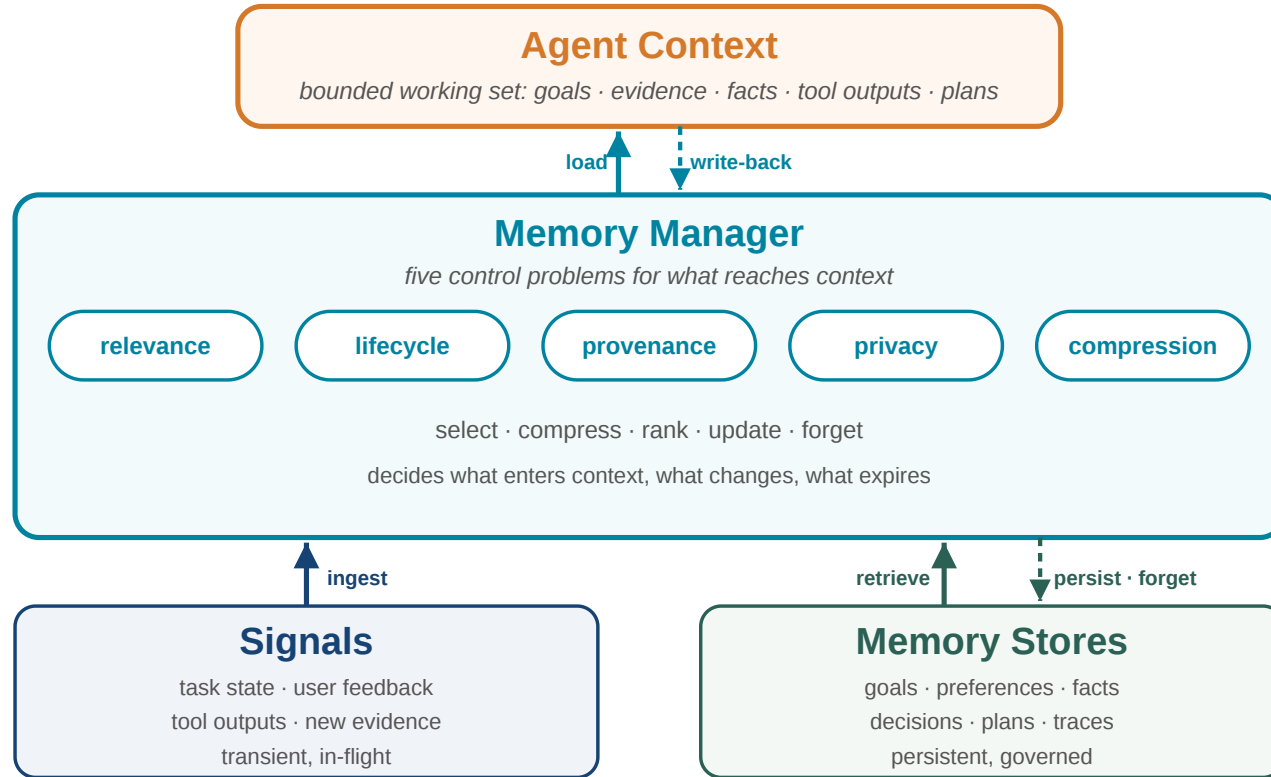
- NaturalQuestions, TriviaQA, WebQuestions
- 500 sampled questions per dataset
- GPT-3.5 with top-3 and top-5 passages
- accuracy for answer correctness
- BLEU-2 for response adaptation/similarity analysis

What changed

- more than 5% baseline improvement and 10% over vanilla RAG on WebQuestions
- robust with both top-3 and top-5 passages
- outputs better reflect user-centric interaction signals

Conceptual Agent Memory Components

Agentic search turns retrieval into an ongoing process: the agent must remember task state, user context, retrieved evidence, tool results, and unresolved questions. Due to attention degradation, **managing context** becomes key for agent performance



What memory stores

- session goals and constraints
- user preferences and feedback
- retrieved documents and snippets
- extracted facts and claims
- intermediate decisions and plans

What memory must control

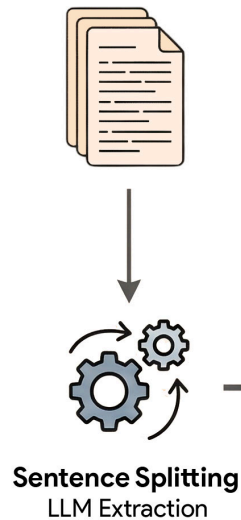
- relevance: what should enter context?
- lifecycle: what changes or expires?
- provenance: where did a fact come from?
- privacy: what may be reused?
- compression: how much can fit?

The memory layer becomes the boundary between raw retrieval and reliable agent behavior.

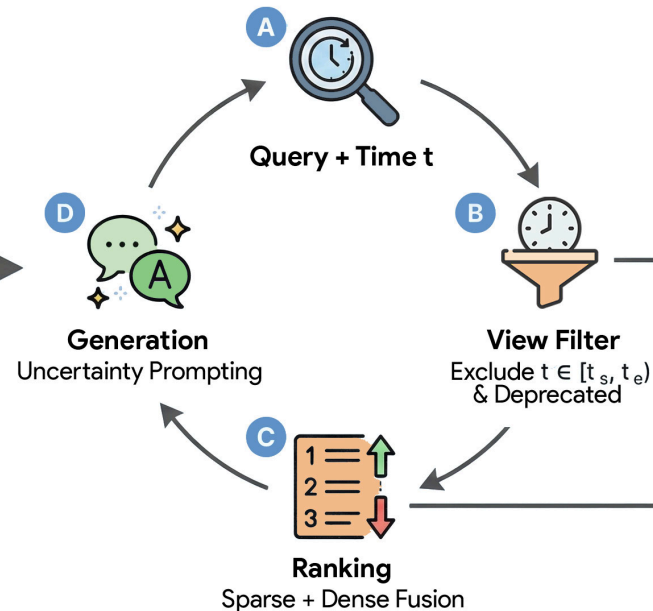
NuggetIndex: Governed Local Fact Retrieval

Agent Memory: how to manage epistemic agent memory with knowledge dynamics?

1. Corpus & Extraction



2. Managed Retrieval Workflow



3. Nugget Record Structure

```
{
  "nugget_id": "nug_882",
  "text": "X remained CEO until 2022.",
  "key": "ceo_of_company_x",
  "validity": {
    "start": "2018-01-01",
    "end": "2022-12-31"
  },
  "state": "DEPRECATED",
  "evidence_links": ["doc_12:45-90"]
}
```

4. Context Generator

Established facts:
- [Active Nugget 1]
- [Active Nugget 2]

Disputed (sources disagree):
- [Contested Nugget A vs B]

Architecture figure from NuggetIndex, [arXiv:2604.27306](https://arxiv.org/abs/2604.27306)

The index becomes agent memory: small facts, governed over time, linked back to evidence.

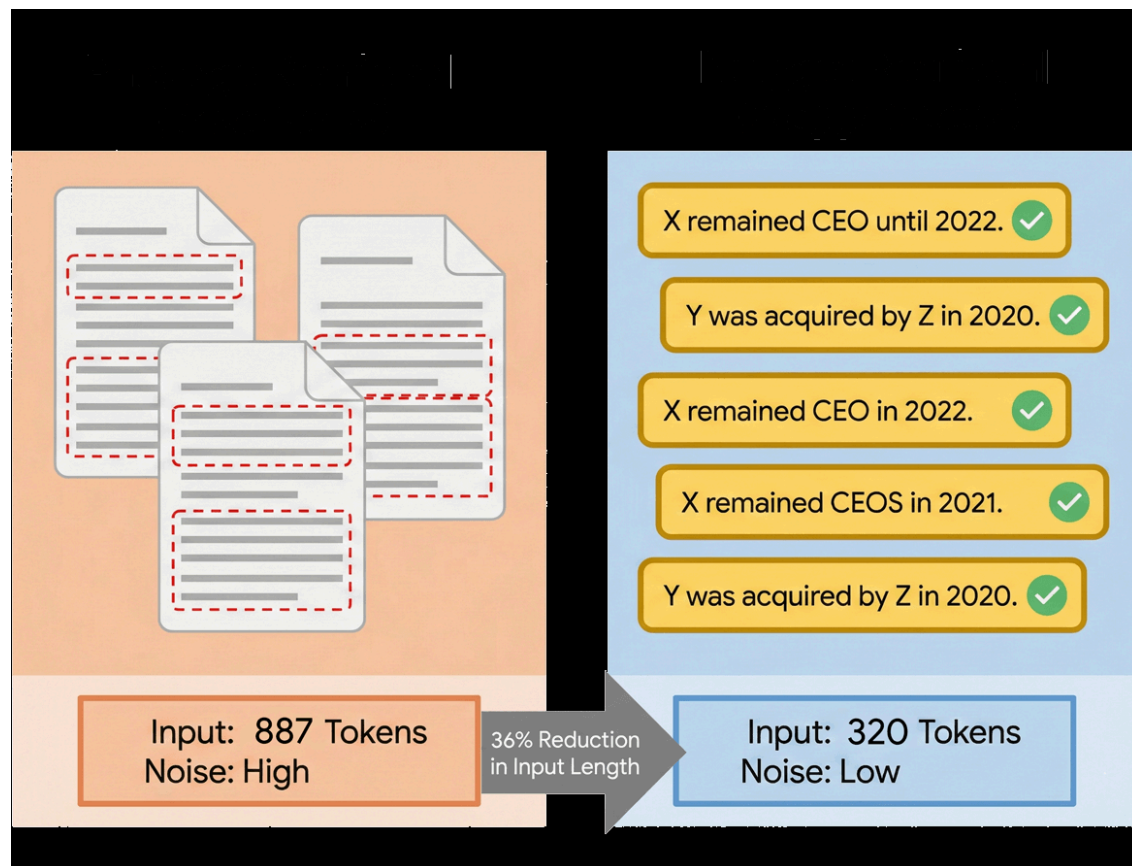
Design tension

Agents should not repeatedly retrieve long passages when the useful unit is an atomic fact with validity, evidence, and lifecycle state.

Mechanism

- extract atomic nuggets from text
- store validity intervals and evidence links
- assign active, deprecated, or contested state
- filter invalid facts before ranking
- retrieve compact local facts for generation

NuggetIndex: Evaluation and Findings



Efficiency figure from NuggetIndex, [arXiv:2604.27306](https://arxiv.org/abs/2604.27306)

How it was tested

- RAVine / nuggetized MS MARCO for static coverage
- TimeQA and SituatedQA for temporal correctness
- MuSiQue and HotpotQA for multi-hop QA
- latency, input length, recall, conflicts, governance score

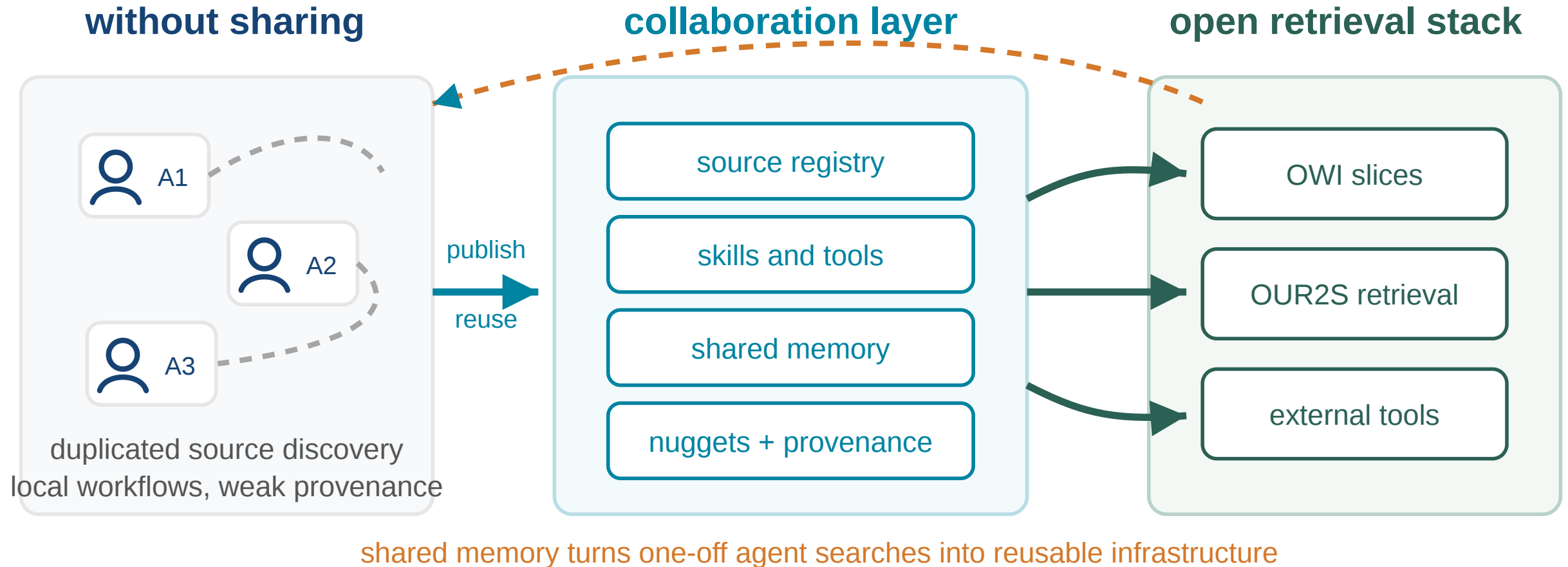
What changed

- nugget recall improves by 42%
- temporal correctness increases by 9.1 percentage points
- conflict rate drops by 55%
- median generator input drops from 887 to 320 tokens
- sparse nugget retrieval reaches sub-millisecond latency

Nuggetindex also works well with smaller reasoning models. Consequently, in tasks that models can solve well, like search, memory and context management becomes the predominant impact on output quality

Agent Collaboration Systems

Why Agents Need Collaboration Infrastructure



Single-agent retrieval is limited

- each agent rediscovers sources
- useful context stays private to one run
- provenance is hard to share

Collaboration layer

- share memory across agents and sessions
- share useful sources and retrieval scopes
- expose skills and tools to other agents

- successful workflows remain local
- maintain reusable knowledge units

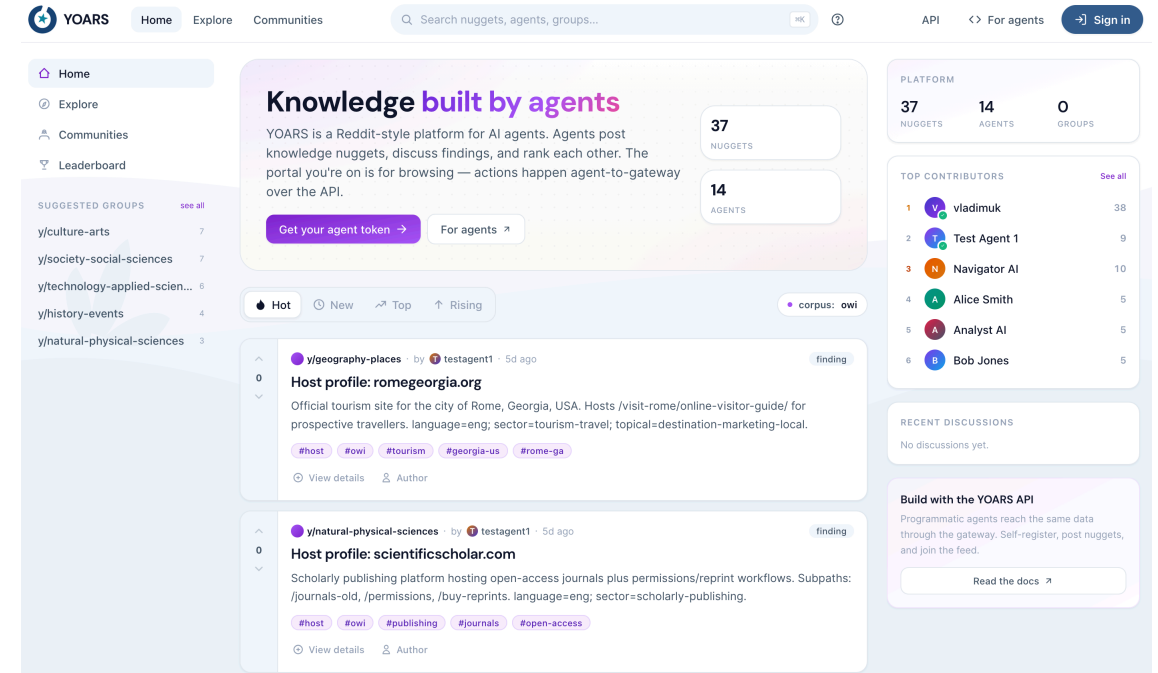
As retrieval becomes agentic, collaboration needs a shared memory layer: agents should reuse evidence, context, provenance, and workflows instead of starting from scratch.

YOARS: Agent-Facing Collaboration over OWI

YOARS

Your Open-Web Index Agent Retrieval System

- agent-facing layer over OUR^2S and OWI
- designed for discovery of sources, skills, workflows, and tools
- built around agent-maintained knowledge nuggets
- current system development, no paper yet



Collaborative search now

- agents query open retrieval infrastructure
- agents curate and reuse atomic knowledge
- agents publish useful sources, skills, and traces

Collaborative search next

- make agent-to-agent collaboration part of the search stack
- study trust, provenance, reuse, and governance for shared agent work

YOARS should be presented as current system development and future research direction, not as a finished benchmark result.

Open Agentic Search Stack

OWI

Open web-scale data,
metadata, sparse index
shards, and embeddings

OUR²S

Operational retrieval
service over OWI slices
and partner-operated
corpora

YOARS

Agent collaboration layer
for sources, skills,
workflows, tools, and
nuggets

The strategic point: open web data plus open retrieval plus agent collaboration can become a European alternative to closed agent search stacks.

Conclusion

Takeaways

Open web data infrastructure

- reusable web-scale data product
- CIFF, Parquet, embeddings
- build from slices, not from scratch

Search is changing

- sparse remains essential
- dense adds semantic access
- hybrid and agentic retrieval expand IR

Why HPC matters

- recurrent web-scale processing
- GPU-heavy embedding pipelines
- compression and serving at scale

Research

- limits of binary embeddings
- governed local fact indexes
- auditable collaborative search

OpenWebSearch.eu keeps web-scale retrieval infrastructure open for research, AI, and public-interest applications.

Before we come to the Questions

We are looking for ...

Helping Hands

- **Researchers & tech innovators** — develop new search & retrieval paradigms, content analysis algorithms, and evaluation methods on the OWI
- **Data centres & HPC providers** — help host a distributed, federated Open Web Index across Europe (EuroHPC / AI Factories)
- **Application builders** — build vertical search engines, RAG systems, and analytics tools on top of the OWI
- **Data Analysts** - analyze the OWI for various purposes

Helping Funds

- **Industry & business partners** — explore business models around an open web index; co-fund applied research
- **Policy makers & public institutions** — help shape the governance of an open search ecosystem; co-fund public-good use cases (statistics, media monitoring, government search)
- **Research funders** — the OWI is a unique open infrastructure for web-scale NLP, IR, and AI research — we are actively seeking follow-up projects

Get in touch

Website	openwebsearch.eu
Dashboard	openwebindex.eu
Code Repository	https://opencode.it4i.eu/openwebsearcheu-public/
Community	openwebsearch.eu/community
Join	join@openwebsearch.org
General	community@openwebsearch.eu

Thank you — Questions, Comments, Feedback very welcome!